

Principles of Economics

Problem Set 5

Universidad Carlos III de Madrid | Department of Economics

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1. WARMING-UP PROBLEMS

- The isoprofit curves for Cheerios are downward sloping, but for Beautiful Cars they slope downward when Q is low and upward when Q is high. In both cases the higher isoprofit curves get closer to the average cost curve as quantity increases. Why?
- What is the reason for the difference in the shape of the isoprofit curves between Cheerios and Beautiful Cars?

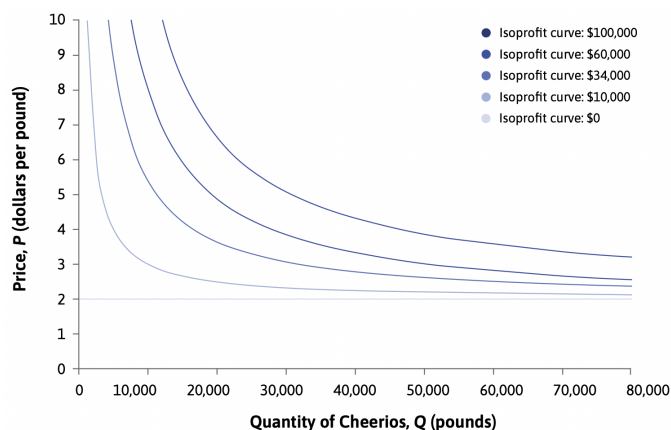


Figure 7.4

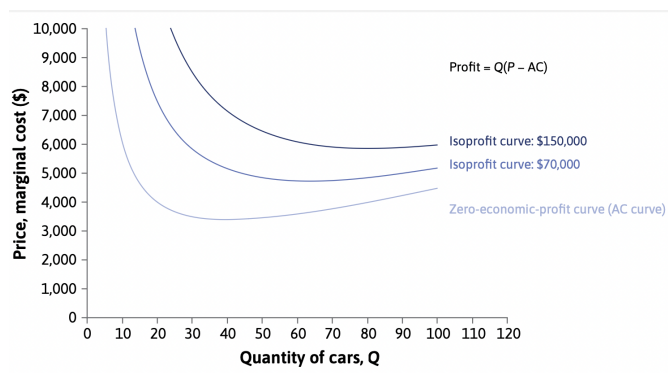


Figure 7.10

- Suppose that Beautiful Cars had sufficient information and so much bargaining power that it could charge each consumer, separately, the maximum they would be willing to pay. Draw the demand and marginal cost curves (as in Figure 7.20) and indicate on your diagram:

- a) the number of cars sold
- b) the highest price paid by any consumer
- c) the lowest price paid
- d) the consumer and producer surplus

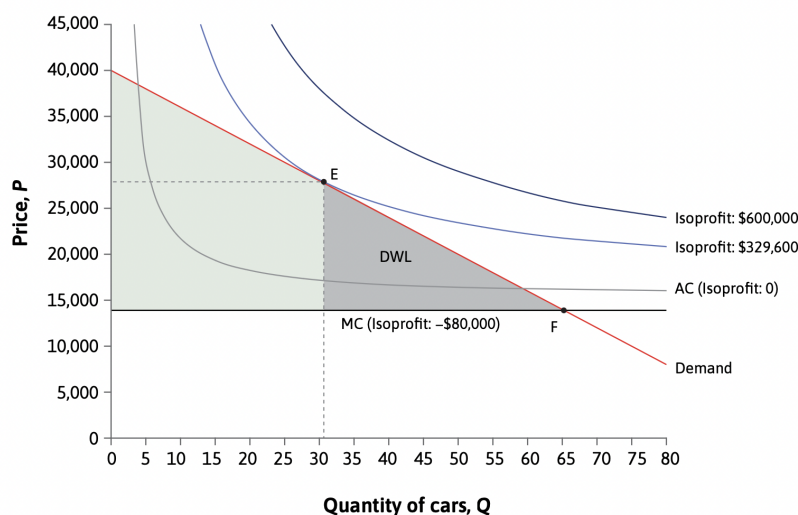


Figure 7.20

- a) Can you think of any examples of goods that are sold in this way?
 - b) Why is this type of pricing not common practice?
 - c) Some firms charge different prices to different groups of consumers, for example airlines may charge higher fares for last-minute travellers. Why would they do this and what effect would it have on the consumer and producer surpluses? What can and cannot we say about the effect on overall welfare in the market?
 - d) Suppose a competition policy is implemented that prevents firms from charging different prices to different groups of consumers. How could this change give consumers more bargaining power?
 - e) Under these rules, how many cars would be sold?
 - f) Under these rules, what would the producer and consumer surpluses be?
4. Below you can see the average and marginal costs per student for the year 1990-1 that Koshal and Koshal calculated from their research.

	STUDENTS	MC (\$)	AC (\$)	TOTAL COST (\$)
Undergraduates	2,750	7,259	7,659	21,062,250
	5,500	6,548	7,348	40,414,000
	8,250	5,838	7,038	
	11,000	5,125	6,727	73,997,000
	13,750	4,417	6,417	88,233,750
	16,500	3,706	6,106	100,749,000
	STUDENTS	MC (\$)	AC (\$)	TOTAL COST (\$)
Graduates	550	6,541	12,140	6,677,000
	1,100	6,821	9,454	10,399,400
	1,650	7,102	8,672	
	2,200	7,383	8,365	18,403,000
	2,750	7,664	8,249	22,684,750
	3,300	7,945	8,228	27,152,400

- How do average costs change as the numbers of students rise?
- Using the data for average costs, fill in the missing figures in the total cost column.
- Plot the marginal and average cost curves for undergraduate education on a graph, with costs on the vertical axis and the number of students on the horizontal axis. On a separate diagram, plot the equivalent graphs for graduates.
- What are the shapes of the total cost functions for undergraduates and graduates? (You could sketch them using what you know about marginal and average costs.) Plot them on a single chart using the numbers in the Total cost column.
- What are the main differences between the universities' cost structures for undergraduates and graduates?
- Can you think of any explanations for the shapes of the graphs you have drawn?

2. MULTIPLE CHOICE QUESTIONS

5. Which of the following events would contribute to economies of scale?

- Large supermarkets being able to drive down the cost of purchasing milk as it increases its scale of business.
- The expansion of the workforce requiring extra layers of management for better performance monitoring.

- c. A merger of two firms leading to large increases in the costs of HR, IT, and legal departments.
- d. The higher the number of users of Windows 10, the less likely it is that Microsoft will be able to achieve a higher sale of its operating system.
6. The following is a table of the total cost (TC) of producing output Q for a particular firm. Based on this information, which of the following statements is correct?

Q	10	20	30	40	50	60	70	80	90	100
$TC (£)$	140	210	265	310	360	420	490	570	660	760

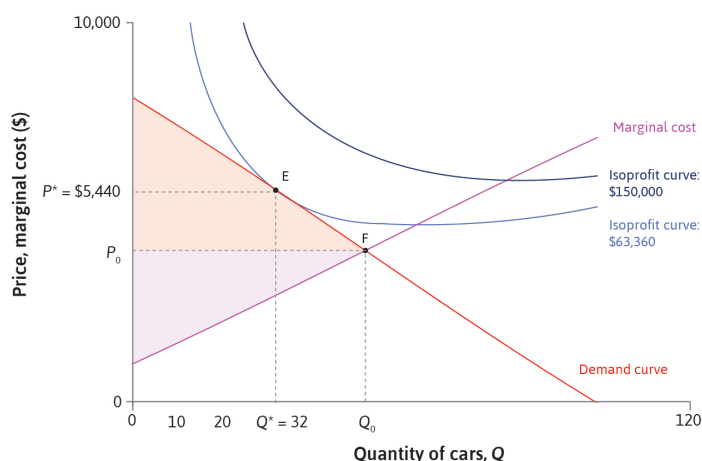
- a. The average cost at $Q = 40$ is £7.
- b. The marginal cost at $Q = 80$ is £9.50.
- c. The marginal cost is higher than the average cost at $Q = 50$.
- d. The marginal cost curve intersects the average cost curve at $Q = 60$.
7. Suppose that the marginal cost of producing a pound of cereal is \$2, irrespective of the level of output, but there are also some fixed costs of production. Which of the following statements is correct?
- a. The total cost curve is an upward-sloping straight line through the origin.
- b. The average cost curve is U-shaped.
- c. The marginal cost curve is a horizontal straight line.
- d. The marginal cost curve intersects the average cost curve at its minimum point.

3. NUMERICAL PROBLEMS

8. A firm's cost of production is £12 per unit of output. P is the price of the output good, and Q is the number of units produced. Using this information, read the following statements and choose the correct option(s).
- a) Point $(Q, P) = (2,000, 20)$ is on the isoprofit curve representing the profit level £20,000.
- b) Point $(Q, P) = (2,000, 20)$ is on a lower isoprofit curve than point $(Q, P) = (1,200, 24)$.
- c) Points $(Q, P) = (2,000, 20)$ and $(4,000, 16)$ are on the same isoprofit curve.
- d) Point $(Q, P) = (5,000, 12)$ is not on any isoprofit curve.

Hint: $[\text{Profit} = (P - \text{unit cost}) \times Q; \text{ in this problem the unit cost is } £12]$

9. The following figure depicts a firm's profit-maximising choice at point E, given the market demand curve and the firm's marginal cost curve. You are given that the firm's marginal costs are \$400, \$2,960 and \$4,200 at output levels $Q = 0$, $Q^* = 32$ (point E) and $Q_0 = 48$ (point F), respectively. Additionally, we know that the demand curve intersects the price and marginal cost axes at \$7920. Based on this information, which of the following statements is correct?



- The consumer surplus at E is \$41,000.
- The producer surplus at E is \$126,720.
- The deadweight loss at E is \$19,840.
- The gains from trade at E are \$120,320.

10. Figure 7.19 shows the spending per week in each category of a US consumer whose total expenditure on food is \$80, with typical spending patterns across food categories.

Category	Type	Calories per serving	Price per 100 g (\$)	Typical spending per week (\$)	Price elasticity of demand
1	Fruit and vegetables	660	0.38	2.00	1.128
2	Fruit and vegetables	140	0.36	3.44	0.830
15	Grain, pasta, bread	1,540	0.38	2.96	0.845
17	Grain, pasta, bread	960	0.53	2.64	0.292
28	Snacks, candy	433	1.13	4.88	0.270
29	Snacks, candy	1,727	0.68	7.60	0.295
30	Milk	2,052	0.09	2.32	1.793
31	Milk	874	0.15	1.44	1.972

Suppose that the price of category 30, high-calorie milk products, increased by 10%:

- By what percentage would his demand for high-calorie milk products fall?
 - Calculate the quantity he consumes, in grams, before and after the price change.
 - Calculate his total expenditure on high-calorie milk products before and after the price change. You should find that expenditure falls.
 - Now choose a category for which the price elasticity is less than 1, and repeat the calculations. In this case you should find that expenditure rises.
- By 17.93%, because $0.1 \times 1.793 = 0.1793$
 - $2.32/0.9 = 2.5778$ kg before, and $2.5778 - (0.1793 \times 2.5778) = 2.1156$ kg after.
 - \$2.32 before, and $2.1156 \times 0.9 \times 1.1 = \2.09 after.

d. Pasta, category 15, elasticity is 0.854. The quantity decreases by 8.54%.

4. CONCEPTUAL QUESTIONS

11. Food taxes intended to shift consumption towards a healthier diet are controversial. Some people think that individuals should make their own choices, and if they prefer unhealthy products, the government should not interfere. In view of the fact that those who become ill will be cared for at some public expense, others argue that the government has a role in keeping people healthy. In your own words, provide arguments for or against food taxes designed to encourage healthy eating.
12. Imagine that you are a politician in a town where a multinational retailer is planning to build a new superstore. A local campaign is protesting that it will drive small independent retailers out of business, and thereby reduce consumer choice and change the character of the area. Supporters of the plan argue in turn that this will only happen if consumers prefer the supermarket. Which side are you on?